ASSESSMENT PORTFOLIO

SC1153 - Introduction to Biotechnology

Batch 13 - June 2026



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Abstract

This assessment portfolio presents my learning journey throughout the SC1153 - Introduction to Biotechnology module. The portfolio documents my participation in four major assignments involving group presentations, biotechnology-based product communication, individual career planning, and an ethical debate.

Assignment 1 introduced me to the environmental applications of biotechnology through the study of Gray Biotechnology, with my individual contribution focusing on Waste-to-Energy and Biofilters. Assignment 2 provided experience in communicating a biotechnology-developed product to the general public through a group video on producing egg white protein using biotechnology.

Assignment 3 focused on career development and allowed me to examine the career journey of Professor Frances H. Arnold and use her experiences to develop my own career trajectory in biotechnology. My proposed career pathway includes developing knowledge in molecular biology, biochemistry, laboratory techniques, protein analysis, bioinformatics and research. These elements are present in my original presentation.

Assignment 4 provided an opportunity to examine the ethical, legal and social dimensions of biotechnology through a debate on Genetic Privacy and Law Enforcement, where I participated as a member of the Opposition team.

Overall, these assignments helped me develop communication, teamwork, presentation, research, critical thinking, creativity, time management and self-reflection skills. The portfolio demonstrates not only the completed academic work but also my personal development as an undergraduate biotechnology student.

Skills developed

Research • Presentation • Teamwork • Communication • Critical thinking • Time management • Creativity • Reflection

Acknowledgements

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of the SC1153 - Introduction to Biotechnology module and the development of this assessment portfolio.

First, I would like to sincerely thank Prof. Colin N. Peiris for providing valuable academic guidance throughout the module. His lectures and guidance on presentation skills, portfolio preparation and academic development helped me understand how to communicate scientific knowledge effectively and professionally. I am also grateful for his involvement in evaluating and assessing the final portfolio.

I would also like to express my appreciation to Mr. Shanitha Mirihana, Module Leader, for his continuous involvement in lectures, assignment guidance and evaluation activities. His constructive guidance and feedback helped me improve my presentation, communication and teamwork skills.

I am thankful to my group members and batch mates for their cooperation, support and contribution throughout the assignments. Working with different members helped me understand the importance of teamwork, responsibility and effective communication.

Finally, I would like to thank my family and everyone who supported me during this academic journey.

Thank you to everyone who contributed to my learning experience and personal development throughout this module.

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Objectives

Assignment 1 – Gray Biotechnology

- **To understand major applications of environmental biotechnology.**
- **To investigate the importance of Waste to Energy and Biofilters.**
- **To develop scientific communication skills.**
- **To contribute effectively to a group presentation.**

The assignment specifically required groups to present applications of their assigned biotechnology area, with each member presenting a section.

Assignment 2 – Biotechnology Product Video

- **To investigate a biotechnology-developed product.**
- **To communicate scientific information to a general audience.**
- **To develop video presentation skills.**
- **To improve teamwork and coordination.**

The guideline required a 10-minute group video with individual contributions from each member.

Assignment 3 – Career Trajectory

- **To study a biotechnology-related role model.**
- **To identify the skills and experiences required for a biotechnology career.**
- **To develop short, medium and long-term career goals.**
- **To connect my career ambitions with a professional role model.**

Assignment 4 – Ethical Debate

- **To understand ethical, legal and social issues in biotechnology.**
- **To develop evidence-based arguments.**
- **To improve critical thinking and public speaking.**
- **To respond to opposing arguments professionally.**

The assigned topic for Group 3 was Genetic Privacy and Law Enforcement.

Assignment 1 - Areas of Biotechnology Applications in the World

Assignment Brief

Assignment One was a group-based academic presentation on a selected area of biotechnology. The purpose of the activity was to develop students' ability to investigate a biotechnology-related area, organize scientific information and communicate the findings effectively to an academic audience.

Our group was assigned Gray Biotechnology, which focuses on the application of biotechnology to environmental protection and sustainable environmental management. The group developed the presentation under the creative title "Nature's Silent Shield".

Group Information

Assigned Area - Gray Biotechnology
Group - 9
My Presentation Position - Presenter 3

Group Members

- Senanayake A.G.M.S - HS26510360
- Devindi J.A.I - HS26510453
- Imesha H.A.Y – HS26510400
- Sarathchandra R.S.S.V – HS26510019

My Role

I participated as the third presenter of the group presentation. My contribution to the group presentation focused on;

1. Waste-to-Energy
2. Biofilters

My contribution

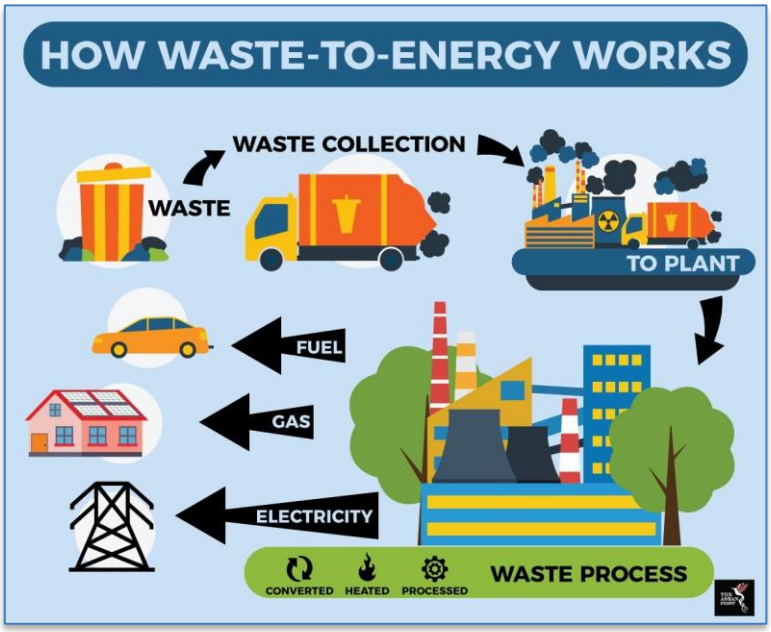
I explained -

01.Waste to Energy

Waste to Energy refers to the conversion of waste materials into useful forms of energy such as electricity, heat or fuel. It is an important approach to sustainable waste management.

Main Methods

- Incineration
- Anaerobic digestion
- Gasification
- Pyrolysis



Advantages

- Reduces waste volume.
- Generates electricity and heat.
- Decreases landfill use.
- Can reduce dependence on fossil fuels.
- May allow recovery of useful materials.

Limitations

- High construction cost.
- Potential air pollution if not properly controlled.
- Requires advanced technology.
- Some processes may generate toxic ash.

02.Biofilters

Biofilters are systems that use microorganisms and natural materials to remove pollutants and unpleasant odors from air or water.

Common Filter Materials

- Compost
- Soil
- Wood chips
- Activated carbon

Working Principle

Polluted air/water → Filter material → Microbial degradation → Cleaner output

Pollutants that can be removed

- Hydrogen sulfide
- Ammonia
- Volatile organic compounds (VOCs)
- Unpleasant odors



Advantages

- Environmentally friendly
- Low operating cost
- Effective odor control
- Simple maintenance
- Reduces harmful emissions

Applications

- Wastewater treatment plants
- Landfills
- Food industries
- Chemical industries
- Agricultural farms

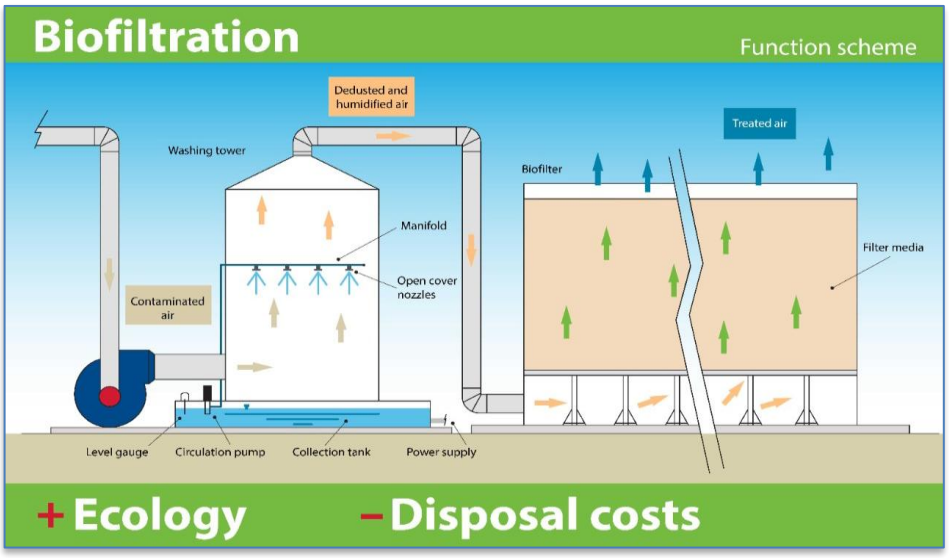




Figure 1.1 - Presentation of the Waste to Energy section during Assignment 1



Figure 1.2 - Presentation of the Biofilters section during Assignment 1.



Figure 1.3 - Group members after the successful completion of the Assignment 01 group presentation.

Assignment 1 - Reflection

Assignment 1 was my first major group presentation experience in this module. Working on Gray Biotechnology helped me understand how biotechnology can be applied beyond laboratories and directly contribute to environmental protection. My specific contribution was Waste to Energy and Biofilters. While preparing these topics, I developed a better understanding of sustainable waste management and biological approaches to pollution control.

The assignment also improved my ability to explain scientific concepts in a simple and understandable manner. Since the presentation was intended to communicate biotechnology applications clearly, I had to focus not only on scientific accuracy but also on how the information was delivered.

Working as Presenter 3 also taught me the importance of maintaining the flow between different speakers. I learned that a successful group presentation depends not only on individual performance but also on coordination among all members.

Skills developed

Through this assignment, I developed -

- **Scientific presentation skills**
- **Public speaking**
- **Teamwork**
- **Research and information-selection skills**
- **Voice projection**
- **Eye contact**
- **Body language**
- **Time management**
- **Ability to explain scientific concepts clearly**

Assignment 1 - Full Presentation

Nature's Silent Shield

Exploring the Applications of Grey Biotechnology

Sequence	Student Name	Registration No.
Presenter 1	Senanayake A.G.M.S	HS26510360
Presenter 2	Devindi J.A.I	HS26510453
Presenter 3	Imesha H.A.Y	HS26510400
Presenter 4	Sarathchandra R.S.S.V	HS26510019

Slide 1

What is Grey Biotechnology ?

The Eco – Guardian

- Grey Biotechnology (also known as Environmental Biotechnology) is a highly specialized branch of biological science.
- Its core directive is developing biological systems to actively combat global environmental cries.

Ex: 1.Cleaning up oil spills in the ocean
2.Using plants to clean polluted soil and water

Key Objectives

1.pollution Abatement : Degrading complex toxic waste into completely harmless compounds.

2.Resource Recovery: Upcycling municipal organic waste into clean green energy (biogas

3.Biodiversity Preservation : Safeguarding natural wildlife and plants from industrial toxicity.

Slide 2

Applications of Grey Biotechnology

1. Phytoremediation
2. Bioremediation
3. Water Purification
4. Waste to energy
5. Biofilters
6. Biosensors



Slide 3

phytoremediation

- The Power of Plants

Direct use of green plants to extract ,stabilize or destroy heavy metals and hazards pollutants from soils.

The Chernobyl Cleanup:

- Following the nuclear disaster, scientists planted sunflowers directly into contaminated lakes.
- Sunflower absorbed highly radioactive elements through their root systems .
- Successfully cleaned and purified

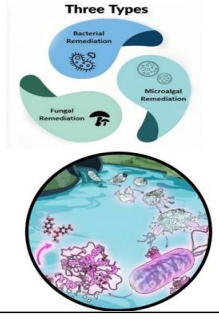


Slide 4

Bioremediation

- Uses microorganisms, plants, fungi, or enzymes
- Breaks down harmful pollutants
- Eco-friendly and cost-effective

Example
- Pseudomonas putida
- Degrades oil spills
- Breaks down hydrocarbons
- Removes toxic pollutants



Slide 5

Water Purification

Special bacteria can

- Remove microplastics
- Absorb heavy metals
  - -Lead
  - -Arsenic
  - -Mercury
- Convert nitrates into nitrogen gas
  - - Denitrification

Benefits

✔ Cleaner water
✔ Healthy aquatic ecosystems
✔ Sustainable wastewater treatment



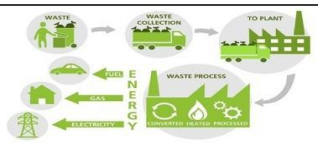
Slide 6

Waste to Energy

What is it?

- Waste-to-energy is the process of converting waste materials into useful energy such as electricity, heat, or fuel.
- It is an important method of sustainable waste management.

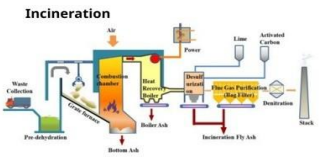




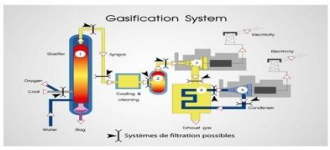
Main Methods

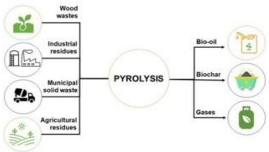
1. Incineration
2. Anaerobic Digestion
3. Gasification
4. Pyrolysis

Slide 7









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Advantages and Disadvantages

Advantages

- Reduces waste volume
- Generates electricity and heat
- Decreases landfill use
- Reduces dependence on fossil fuels
- Can recover useful materials from waste

Disadvantages

- High construction cost
- Air pollution if not properly controlled
- Requires advanced technology
- Some methods produce toxic ash



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Biofilters

What is it?

- Biofilters are systems that use microorganisms and natural materials to remove pollutants and odors from air or water.

Common natural filter materials

- ✓ Compost
- ✓ Soil
- ✓ Wood chips
- ✓ Activated carbon



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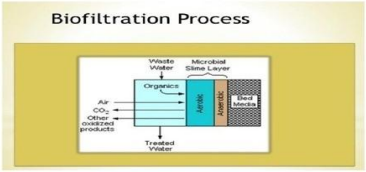
How Biofilters Work

- Working Principle

1. Polluted air or water enters the biofilter.
2. Pollutants are trapped in the filter material.
3. Microorganisms break down harmful substances into simpler harmless compounds.
4. Clean air or water exits the system.

Pollutants Removed

- *Hydrogen sulfide*
- *Ammonia*
- *Volatile organic compounds (VOCs)*
- *Bad odors*



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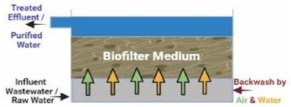
Advantages and Applications

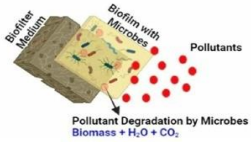
Advantages

- Eco-friendly technology
- Low operating cost
- Effective odor control
- Simple maintenance
- Reduces harmful emissions

Applications

1. Wastewater treatment plants
2. Landfills
3. Food industries
4. Chemical industries
5. Agricultural farms

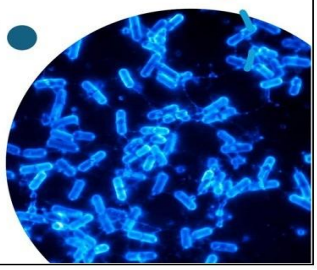




Slide 12

"Glowing Alerts" (Biosensors)

- Using bioluminescence bacteria to detect heavy metals.
- GM bacteria such as,
  - Engineered E. coli
  - Pseudomonas putida
  - Bacillus subtilis
- Genetically modified by Inserting metal responsive promoter and lux reporter gene
- more heavy metal = more bioluminescence produced



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In near future,

- **Self - healing biocities**
  - self healing concrete
  - water purification
  - waste recycling
  - air purification
- **GMO**
  Bacteria and fungi
  - Heavy metal remover
  - Plastic decomposition
  - Carbon capture



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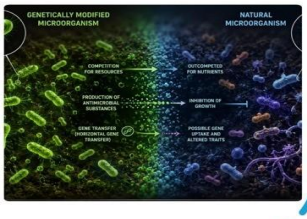
Concerns....

Environmental concerns

- GMO competing with native species
- Horizontal gene transferring
- Unpredictable ecological effects

Social concerns

- Ethical issues
- Regulatory challanges
- High cost
- Public acceptance



Slide 15

Healer of Mother Nature

Grey biotechnology is all about healing earth using the same components earth offers.

Enhancing microorganisms, plants and bio processes is the technological or human aspect of it.

Those modified products are used to heal the earth by cleaning air, water and soil.

Taking necessary risk assessment, implementing regulations and educating public while releasing grey biotechnology is the way to a safer, cleaner and more sustainable world.



Slide 16

References

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Slide 17

Presentation Slides : Click Here to View

Assignment 2 - Introducing an Exciting New Product Developed Using Biotechnology

Assignment brief

Assignment Two was a group-based educational video assignment designed to introduce an interesting biotechnology product to a general audience.

The task required the group to develop approximately a 10-minute video, including an introduction, main content and conclusion. Each group member was required to present an individual section before the sections were combined into the final video.

Group Information

Topic - Producing Egg White Protein Using Biotechnology
Group - 4

Group Members

- Sathsarani E.S.P.S - HS26510335
- Bandaranayake B.M.M.S - HS26510338
- Imesha H.A.Y - HS26510400
- Selitha L.D – HS26510293

Video evidence

The submitted group video was used as evidence. Selected frames below document the video content and individual presentation segments.



Video frame 1



Video frame 2



Video frame 3



Video frame 4



Video frame 5



Video frame 6



Video frame 7

[Watch Assignment 02 Video](#) (click here)

MY VIDEO CONTRIBUTION

My sections focused on -

"Advantages of Producing Egg White Protein Using Biotechnology and Future Benefits." (Video frame 4 and 6)

Advantages of Biotechnology-Based Egg White Protein Production

My section focused on how biotechnology-based production can provide potential advantages compared with relying only on conventional egg production.

The discussion highlighted areas such as -

- Potential for controlled protein production.
- Potential reduction of dependence on conventional egg sources.
- Opportunities for consistent product quality.
- Possibility of scalable production.
- Potential applications in food and biotechnology industries.

Future Benefits

Biotechnology-based protein production may provide opportunities for -

- *Alternative protein production systems.*
- *More controlled manufacturing processes.*
- *Development of specialized protein products.*
- *Further innovation in food biotechnology.*
- *Expansion of sustainable biotechnology applications.*



Figure 3.1



Figure 3.2

Figure 3.1/3.2 - Group members during the preparation and recording of the Assignment 03 video presentation.



Figure 3.4 - Receiving valuable feedback and comments from Mr. Shanitha Mirihana following the completion of the Assignment 03 video presentation



Figure 3.2 - Group members after the successful completion of the Assignment 03 video presentation.

Assignment 2 - Summary & Reflection

Assignment 2 provided an opportunity to explore how biotechnology can be used to develop innovative products for public and industrial applications. Our group selected the topic of producing egg white protein using biotechnology and developed a video presentation for a general audience.

My responsibility was to explain the *advantages of producing egg white protein using biotechnology and its potential future benefits*.

Through this task, I learned that scientific information must be communicated in a way that is understandable to people without a biotechnology background. Preparing my section required me to identify the key ideas and present them clearly rather than simply providing complex scientific information.

The group video also required coordination between individual contributions. Each member had to prepare and present their own section, after which the different sections were combined into one complete presentation.

Reflection

Assignment 2 was different from Assignment 1 because the main deliverable was a video rather than a conventional classroom presentation.

This experience helped me develop my ability to communicate scientific information through a digital medium. I became more aware of voice clarity, presentation style, body language and the importance of maintaining audience interest.

I also learned that producing a successful group video requires planning and coordination. Each individual section needed to connect naturally with the next section to maintain the overall flow.

Skills Developed

- Video presentation skills
- Scientific communication
- Teamwork
- Content preparation
- Verbal communication
- Camera awareness
- Voice clarity
- Timing
- Collaboration
- Creativity

Assignment 3 - Career Trajectory in Biotechnology

Assignment brief

Assignment Three was an individual presentation focused on developing a personal career trajectory in biotechnology.

The task required me to research a biotechnology-related role model, investigate the person's educational and professional journey, identify important career milestones and develop a realistic personal career pathway with short-term, medium-term and long-term goals

Student Information

Name – Imesha H.A.Y
Student ID - HS26510400
Assignment Type - Individual Presentation

Selected Role Model

Professor Frances H. Arnold

Why I Selected Her

I selected Frances Arnold because her career demonstrates how interdisciplinary knowledge, research, innovation, persistence and scientific curiosity can contribute to major developments in biotechnology.



Presentation Content

My presentation examined -

- **Role model research**
- **Career achievements**
- **Directed evolution**
- **Contributions to biotechnology**
- **Lessons learned**
- **My career trajectory**
- **Short-term goals**
- **Medium-term goals**
- **Long-term goals**
- **Alignment with my role model**
- **Personal reflection**

My Career Trajectory

1. Short-Term Goals (2026–2027)

My immediate goals include strengthening my knowledge in -

- **Molecular Biology**
- **Biochemistry**
- **Genetics**
- **Analytical Chemistry**



I also aim to develop practical skills in -

- **Laboratory techniques**
- **Protein analysis**
- **Spectroscopy**
- **Chromatography**
- **Python**
- **Bioinformatics**

I plan to gain practical exposure through laboratory activities, research activities, workshops and biotechnology-related internship opportunities.

2. Medium-Term Goals (2027–2029)

My medium-term goals include -

- **Completing the BSc Hons in Biotechnology**
- **Completing an undergraduate research project**
- **Gaining laboratory and research experience**
- **Obtaining biotechnology internship experience**
- **Developing stronger knowledge in protein and enzyme biotechnology**
- **Improving bioinformatics and data-analysis skills**

3. Long-Term Goals – 2030+

My long-term career pathway is to progress towards a biotechnology research career through -

- **Further research experience**
- **Postgraduate education**
- **MSc/PhD studies**
- **Research scientist positions**
- **Progression towards senior biotechnology research responsibilities**

Connection with My Role Model

Professor Frances H. Arnold's journey helped me understand the importance of combining scientific knowledge with research experience, innovation and continuous learning.

I connected her career pathway with my own planned development in biotechnology, particularly in molecular biology, biochemistry, laboratory work, protein/enzyme biotechnology, research and postgraduate education.





Assignment 3 - Reflection

Before completing this assignment, I mainly viewed a biotechnology career as completing a degree and obtaining a job. Researching Professor Frances H. Arnold helped me understand that professional growth also requires research experience, interdisciplinary knowledge, innovation, persistence, collaboration and continuous learning.

The most valuable part of this assignment was converting a general career interest into a structured plan. Dividing my goals into short, medium and long-term stages helped me identify the specific knowledge and skills I need to develop.

Personal learning formula

Knowledge + Skills + Research + Persistence = Career Growth

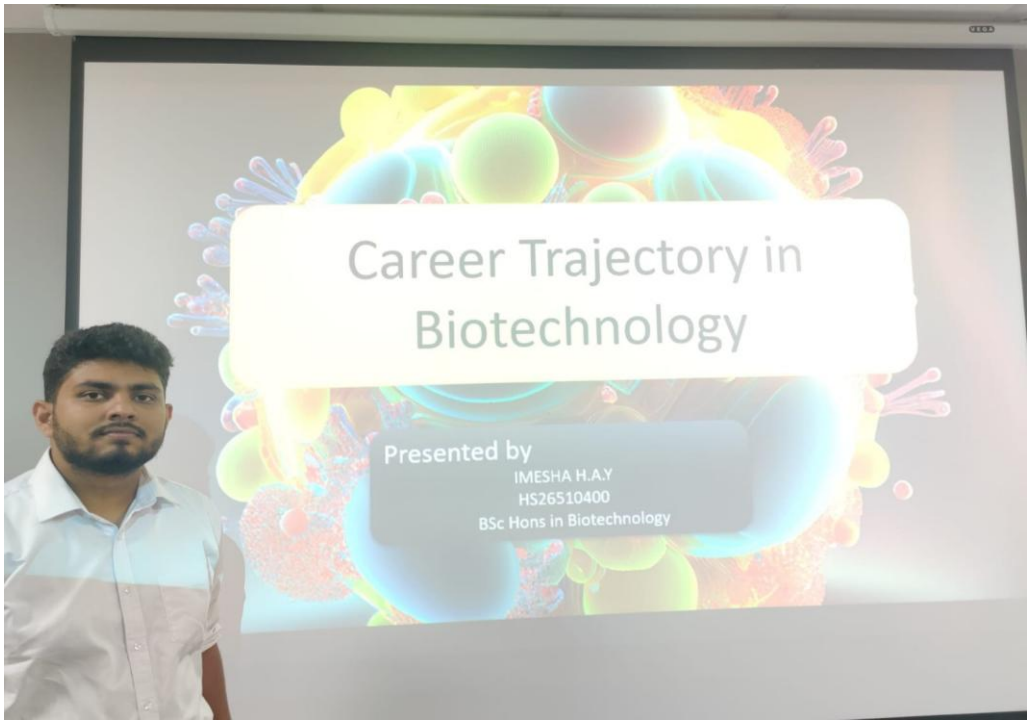
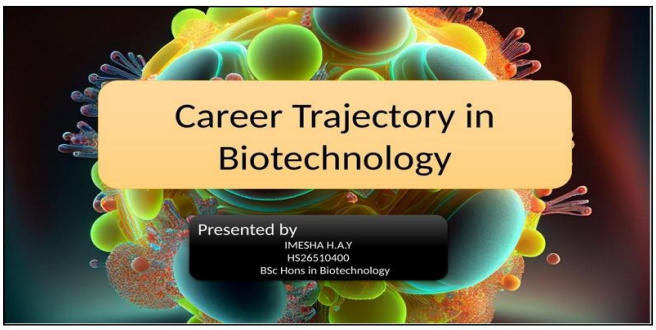


Figure 3.1 - Individual presentation on “Career Trajectory in Biotechnology”.



Figure 3.2 - Presenting my career trajectory and goals.

Assignment 3 - Presentation Handout



Slide 1



Slide 2

Who is Frances Arnold?



- Born - 25 July 1956
- American chemical engineer and researcher
- B.S. Mechanical & Aerospace Engineering - Princeton University, 1979
- Ph.D. Chemical Engineering - University of California, Berkeley, 1985
- Joined Caltech as a visiting associate in 1986
- Became a Caltech professor
- Nobel Prize-winning scientist 2018
- Pioneer in directed evolution
- Major contribution to enzyme engineering





Research areas

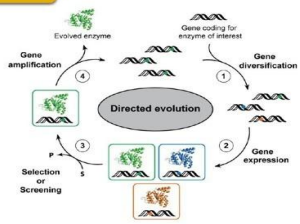
- Protein engineering
- Directed evolution
- Synthetic biology
- Biocatalysis
- Biofuels

Slide 3

Directed Evolution of Enzymes

Frances Arnold's most important contribution is directed evolution. Instead of trying to completely design an enzyme from scratch, scientists generate many variants of an enzyme, test them and select the variants with useful properties. The best variants can then be modified and tested again.

By repeating this process, scientists can obtain enzymes with improved or new functions. This approach uses the principles of evolution as a tool for biotechnology.



Slide 4

Contributions to Biotechnology

Why her work matters to biotechnology

- **Enzyme Engineering**
  Developing enzymes with useful properties.
- **Industrial Biotechnology**
  Using biological catalysts in industrial processes.
- **Pharmaceutical Biotechnology**
  Enzymes can support the production of pharmaceutical compounds.
- **Sustainable Biotechnology**
  Biological catalysts can help develop more environmentally friendly processes.
- **Biofuels**
  Engineered enzymes can contribute to biomass conversion and biofuel research.
- **Synthetic Biology**
  Engineering biological systems for useful functions.

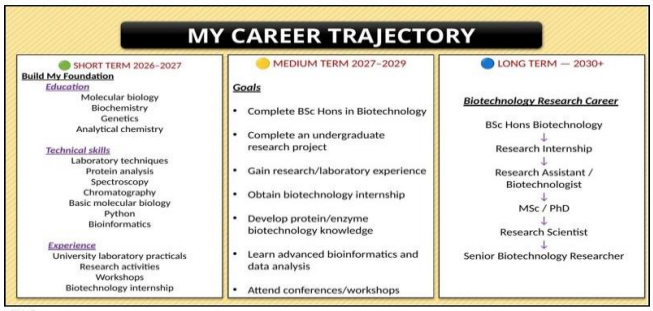
Slide 5

What I Learned From My Role Model

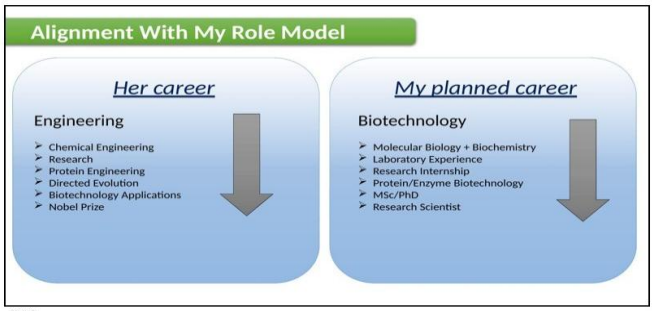
- Lessons from Frances Arnold

What she demonstrated	What I need to develop
Interdisciplinary knowledge	Biology + chemistry + technology
Research	Practical research experience
Innovation	Creative problem-solving
Persistence	Ability to learn from failure
Collaboration	Teamwork and communication
Scientific curiosity	Continuous learning

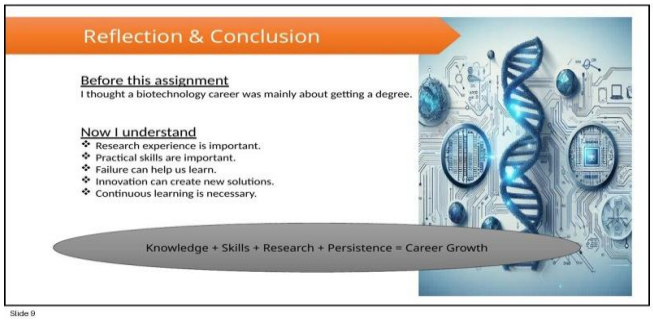
Slide 6



Slide 7



Slide 8



Slide 9



Slide 10

Presentation Slides: Click Here to View

Assignment 4 - Debate on Ethical Issues in Biotechnology

Assignment brief

The task required groups to debate an ethical issue in biotechnology. Group 3 was assigned Genetic Privacy and Law Enforcement: whether law enforcement should be allowed to use consumer DNA databases to track suspects through their relatives.

Group Information

Group – 3

Group Members

Affirmative team	Opposition team.
Punchihewa H.G.I - HS26510283	**Nagahawatta B.I.D – HS26200917**
Selitha L.D - HS26510293	**Herath H.M.N.Y – HS26510280**
Sathsarani E.S.P.S - HS26510335	**Imesha H.A.Y - HS26510400**

My Team - Opposition / Negative
My Position - Third Speaker

Debating Theme - Genetic Privacy and Law Enforcement

Debate Structure

- Round 1 - Main arguments
- Round 2 - Counter arguments
- Final statements

The guideline specifies separate affirmative and opposing teams and provides two-minute speaking allocations for the debate rounds.

Our group was divided into affirmative and opposing teams. I participated on the Opposition team as the third speaker.

MY DEBATE CONTRIBUTION

My Role as Opposition Speaker

Round 01 - Main Opposition Arguments

My role was to present arguments concerning the privacy, security and long-term risks associated with the use of commercial genetic databases

1. Private Ownership of Genetic Databases

Commercial DNA databases are controlled by private companies, meaning their policies and conditions can potentially change over time.

2. Data Breach Risks

Genetic information is highly sensitive, and unauthorized access or data breaches can create serious privacy concerns.

3. DNA Is Not Like a Password

A password can be changed after it is compromised. Genetic information is fundamentally different because an individual's DNA cannot simply be replaced after a data leak.

4. Future Changes in Data Access

Changes in company ownership, policies or access arrangements could affect how genetic information is used.

5. Mission Creep

Information collected for one purpose could potentially be used for additional purposes beyond the original intention.

6. Transparency

Private systems may not provide the same level of transparency expected from public legal processes.

These points formed the main technology and data-privacy focus of my Round 01 contribution.

Round 02 - Counterarguments

In Round 02, I responded to the arguments presented by the Affirmative team.

I acknowledged that investigative genetic genealogy can assist in solving serious crimes, but argued that usefulness alone does not eliminate privacy concerns.

My response focused on -

- **Privacy of relatives**
- **Lack of consent from individuals indirectly affected**
- **Permanent nature of genetic information**
- **Changing company policies**
- **Limitations of using successful cases as proof of complete safety**
- **Difference between a DNA match and proof of guilt**
- **Need for appropriate safeguards**

Key Debate Message

"I am not against DNA technology; I am against uncontrolled access to private DNA data."



Figure 4.1 - Presenting my arguments as the third speaker of the Opposition team during Round 01



Figure 4.2 - Presenting my counterarguments during the second round of the debate.

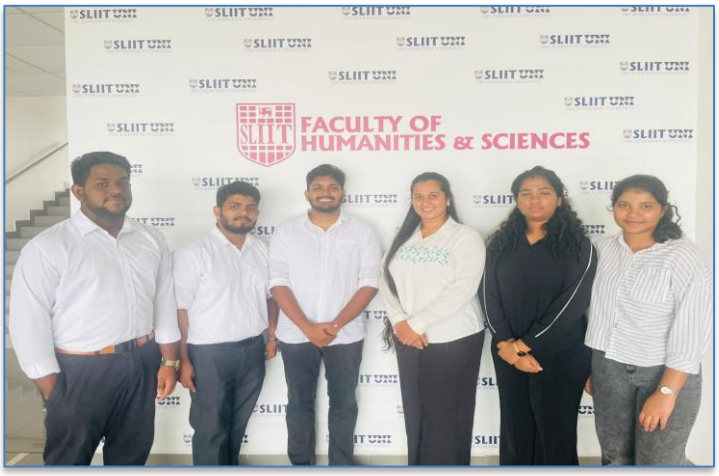


Figure 4.3



Figure 4.4

Figure 4.3/4.4 - Group photograph following the completion of the ethical debate.

Assignment 4 - Reflection

This debate was different from the earlier presentation tasks because I had to listen carefully to another team and respond directly to their arguments. I learned that an effective debate argument should not only state a position; it should explain the reason, recognize the strongest part of the opposing argument and then show why concerns still remain.

The topic also helped me understand that biotechnology can create benefits while simultaneously creating ethical, legal and social questions. The discussion was not simply about whether DNA technology works; it was also about privacy, consent, data ownership, transparency and safeguards. Preparing two rounds of speaking improved my ability to organize ideas quickly and speak persuasively while remaining respectful of the opposing team.

Skills Developed

- Critical thinking
- Argument development
- Rebuttal
- Active listening
- Persuasive communication
- Public speaking
- Ethical reasoning
- Team coordination
- Evidence-based discussion
- Confidence under pressure

Overall Learning Journey

Skill / area	Evidence from my learning
Research & scientific understanding	Selecting relevant information for environmental biotechnology, biotechnology products, career research and ethical debate.
Presentation & communication	Explaining scientific concepts, presenting individually, recording video content and responding in a debate.
Teamwork	Coordinating with group members, maintaining flow and contributing reliably to shared outcomes.
Critical thinking	Connecting evidence with personal goals and evaluating opposing arguments.
Professional development	Thinking about skills, experience, communication and career planning alongside academic knowledge.
Creativity	Using slides, video, visual evidence and structured portfolio design to communicate learning.

My growth across the module

Communicating science clearly → working effectively with others → planning my career → defending an evidence-based position.

Challenges Faced and How I Overcame Them

Assignment 01 – Group Presentation: Gray Biotechnology

One of the main challenges I faced during Assignment 01 was speaking confidently in front of an audience and explaining technical biotechnology concepts in a clear and understandable way. Since my sections focused on Waste-to-Energy and Biofilters, I had to understand the concepts well and explain them within the limited presentation time. I initially felt nervous about presenting and maintaining a proper flow while speaking. To overcome this challenge, I prepared my content carefully, practiced my section several times, and focused on understanding the concepts rather than memorizing every sentence. I also coordinated with my group members to ensure that our presentation had a smooth flow. This experience helped me improve my public speaking, teamwork, time management, and presentation skills.

Assignment 02 – Educational Video: Producing Egg White Protein Using Biotechnology

The main challenge I faced in Assignment 02 was working as part of a group to produce an educational video while managing the different tasks involved in recording and presenting the content. Unlike a normal classroom presentation, a video required preparation, suitable recording conditions, coordination among group members, and clear communication. I had to make sure that my individual section on the advantages and future benefits of producing egg white protein using biotechnology was accurate and easy to understand. I overcame these difficulties by preparing my part in advance, practicing before recording, communicating with my group members, and being flexible with the recording process. Through this assignment, I learned how to work collaboratively, communicate effectively, adapt to unexpected situations, and present scientific information in a more engaging way.

Assignment 03 – Individual Presentation: Career Trajectory in Biotechnology

The biggest challenge I faced in Assignment 03 was preparing and delivering an individual presentation without relying on group members for support. I had to independently research my role model, Professor Frances H. Arnold, understand her contribution to biotechnology, and connect her career journey with my own future career goals. I also found it challenging to organize a large amount of information into a clear presentation while managing the time limit. To overcome these challenges, I planned my presentation carefully, selected only the most relevant information, organized my ideas logically, and practiced my speech several times. This assignment increased my self-confidence, research ability, independent learning, communication skills, and ability to plan my future career path.

Assignment 04 – Debate: Genetic Privacy and Law Enforcement

The major challenge I faced in Assignment 04 was participating in a formal debate on a complex and sensitive topic while presenting arguments clearly and responding to opposing views. As the third speaker of the Opposition team, I had to discuss technology and data privacy, including issues related to commercial DNA databases, data breaches, privacy, and access to genetic information. I needed to understand both sides of the argument so that I could present strong points and respond effectively during the debate. I overcame this challenge by researching the topic, preparing my arguments in advance, practicing my speech, and working closely with my team members to maintain a consistent argument. This experience helped me develop critical thinking, argumentation, teamwork, public speaking, active listening, and the ability to communicate confidently under pressure.

Conclusion

The SC1153 - Introduction to Biotechnology module provided me with a broader understanding of biotechnology and its applications beyond theoretical knowledge.

Through the four assignments, I experienced different forms of academic and professional communication. Gray Biotechnology introduced me to environmental applications such as Waste-to-Energy and Biofilters. The biotechnology product video helped me develop digital communication skills. The Career Trajectory assignment encouraged me to think seriously about my future professional development, while the ethical debate challenged me to analyze biotechnology from legal, social and ethical perspectives.

One of the most important lessons I gained from this module is that a biotechnology professional requires more than scientific knowledge. *Communication, teamwork, critical thinking, creativity, research ability, adaptability and continuous learning* are equally important.

I now have a clearer understanding of the skills and experiences I need to develop throughout my undergraduate journey. My goal is to continue strengthening my scientific knowledge, laboratory skills, research experience and communication abilities so that I can progress towards a successful career in biotechnology research.

References

This portfolio follows the supplied SC1153 Assessment Portfolio template and the Batch 13 assignment guidelines. The presentation handouts are reproduced from the supplied PowerPoint files in the required 6-slides-per-page format.

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